

SC3103: Embedded Programming

Tutorial 3: PCP/SRP

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1. Consider a periodic task system defined as follows.

- Tasks $\tau_1 = (7, 3, 7)$, $\tau_2 = (11, 2, 11)$ and $\tau_3 = (13, 4, 13)$, where for a task $\tau_i = (T_i, C_i, D_i)$, T_i is release period, C_i is worst-case execution time and D_i is relative deadline.
 - Suppose there are two shared resources R_1 and R_2 such that R_1 is used by τ_1 and τ_2 and R_2 is used by τ_2 and τ_3 .
 - Assume that each task will execute for 1 time unit and request for all the shared resources at the end of that time unit (R_1 first, then R_2 if requesting both). It will then hold the resources until it completes.
 - Suppose the first job of τ_3 is released at time 0, that of τ_2 is released at time 1 and that of τ_1 is released at time 3.
- (a) Construct the schedule for the above system under **Rate Monotonic (RM) with PCP** for the time period $[0, 10]$. In your schedule, explain the decisions made by PCP.
- (b) Construct the schedule for the above system under **Earliest Deadline First (EDF) scheduling with SRP** for the time period $[0, 10]$. In your schedule, explain the decisions made by SRP.